

6	(a) two waves travelling (along the same line) in opposite directions overlap/meet same frequency / wavelength resultant displacement is the sum of displacements of each wave / produces nodes and antinodes	M1 A1	B1	[3]
	(b) apparatus: source of sound + detector + reflection system adjustment to apparatus to set up standing waves – how recognised measurements made to obtain wavelength	B1 B1 B1		[3]
	(c) (i) at least two nodes and two antinodes	A1		[1]
	(ii) node to node = $\lambda / 2 = 34 \text{ cm}$ (allow 33 to 35 cm) $c = f\lambda$ $f = 340 / 0.68 = 500$ (490 to 520) Hz	C1 C1 A1		[3]

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7 (a) $W = 1$ and $X = 0$

A1 [1]